

EEE4114F: Problem Set 1 – Solutions

1. (a) $T(x[n]) = \sum_{k=n_0}^n x[k]$

- Not Stable: $|x[n]| \leq M \rightarrow |T(x[n])| \leq \sum_{k=n_0}^n |x[k]| \leq (n - n_0 + 1)M$. As $n \rightarrow \infty$, $T \rightarrow \infty$, so not stable.
- Not Causal: $T(x[n])$ depends on the future values of $x[n]$ when $n < n_0$, so this is not causal.
- Linear:

$$\begin{aligned} T(ax_1[n] + bx_2[n]) &= \sum_{k=n_0}^n ax_1[k] + bx_2[k] \\ &= a \sum_{k=n_0}^n x_1[k] + b \sum_{k=n_0}^n x_2[k] \\ &= aT(x_1[n]) + bT(x_2[n]) \end{aligned}$$

The system is linear.

- Not TI:

$$\begin{aligned} T(x[n - n_0]) &= \sum_{k=n_0}^n x[k - n_0] \\ &= \sum_{k=0}^{n-n_0} x[k] \\ &\neq y[n - n_0] = \sum_{k=n_0}^{n-n_0} x[k] \end{aligned}$$

The system is not TI.

- Not Memoryless: Values of $y[n]$ depend on past values for $n > n_0$, so this is not memoryless.

(b) $T(x[n]) = x[n - n_0]$

- Stable: $|T(x[n])| = |x[n - n_0]| \leq M$ if $|x[n]| \leq M$, so stable.
- Causality: If $n_0 \geq 0$, this is causal, otherwise it is not causal.
- Linear:

$$\begin{aligned} T(ax_1[n] + bx_2[n]) &= ax_1[n - n_0] + bx_2[n - n_0] \\ &= aT(x_1[n]) + bT(x_2[n]) \end{aligned}$$

This is linear.

- TI: $T(x[n - n_d]) = x[n - n_0 - n_d] = y[n - n_d]$. This is TI.
- Not memoryless: Unless $n_0 = 0$, this is not memoryless.

(c) $T(x[n]) = e^{x[n]}$

- Stable: $|x[n]| \leq M$, $|T(x[n])| = |e^{x[n]}| \leq e^{|x[n]|} \leq e^M$, this is stable.
- Causal: It doesn't use future values of $x[n]$, so it causal.
- Not linear:

$$\begin{aligned} T(ax_1[n] + bx_2[n]) &= e^{ax_1[n] + bx_2[n]} \\ &= e^{ax_1[n]} e^{bx_2[n]} \\ &\neq aT(x_1[n]) + bT(x_2[n]) \end{aligned}$$

This is not linear.

- TI: $T(x[n - n_0]) = e^{x[n - n_0]} = y[n - n_0]$, so this is TI.
- Memoryless: $y[n]$ depends on the n^{th} value of x only, so it is memoryless.

(d) $T(x[n]) = ax[n] + b$

- Stable: $|T(x[n])| = |ax[n] + b| \leq a|M| + |b|$, which is stable for finite a and b .
- Causal: This doesn't use future values of $x[n]$, so it is causal.
- Not linear:

$$\begin{aligned} T(cx_1[n] + dx_2[n]) &= acx_1[n] + adx_2[n] + b \\ &\neq cT(x_1[n]) + dT(x_2[n]) \end{aligned}$$

This is not linear.

- TI: $T(x[n - n_0]) = ax[n - n_0] + b = y[n - n_0]$. It is TI.
- Memoryless: $y[n]$ depends on the n^{th} value of $x[n]$ only, so it is memoryless.

(e) $T(x[n]) = x[-n]$

- Stable: $|T(x[n])| = |x[-n]| \leq M$, so it is stable.
- Not causal: For $n < 0$, it depends on the future value of $x[n]$, so it is not causal.
- Linear:

$$\begin{aligned} T(ax_1[n] + bx_2[n]) &= ax_1[-n] + bx_2[-n] \\ &= aT(x_1[n]) + bT(x_2[n]) \end{aligned}$$

This is linear.

- Not TI:

$$\begin{aligned} T(x[n - n_0]) &= x[-n - n_0] \\ &\neq y[n - n_0] = x[-n + n_0] \end{aligned}$$

This is not TI.

- Not memoryless: For $n \neq 0$, it depends on a value of x other than the n^{th} value, so it is not memoryless.

(f) $T(x[n]) = x[n] + u[n + 1]$

- Stable: $|T(x[n])| \leq M + 3$ for $n \geq -1$ and $|T(x[n])| \leq M$ for $n < -1$, so it is stable.
- Causal: Since it doesn't use future values of $x[n]$, it is causal.
- Not linear:

$$\begin{aligned} T(ax_1[n] + bx_2[n]) &= ax_1[n] + bx_2[n] + 3u[n + 1] \\ &\neq aT(x_1[n]) + bT(x_2[n]) \end{aligned}$$

This is not linear.

- Not TI:

$$\begin{aligned} T(x[n - n_0]) &= x[n - n_0] + 3u[n + 1] \\ &= y[n - n_0] \\ &= x[n - n_0] + 3u[n - n_0 + 1] \end{aligned}$$

This is not TI.

- Memoryless: $y[n]$ depends on the n^{th} value of x only, so this is memoryless.

2. (a) $x[n]$ is periodic with period 8:

$$e^{j(\frac{3\pi}{4}n)} = e^{j(\frac{3\pi}{4})(n+N)} = e^{j(\frac{3\pi}{4}n+2\pi k)}$$

$$\implies 2\pi k = \frac{3\pi}{4}N, \text{ for integers } k, N$$

$$\implies N = \frac{8}{3}k, \text{ for integers } k, N$$

The smallest k for which both k and N are integers is 3, resulting in the period N being 8.

(b) $x[n] = [\sin(\pi n/5)]/(\pi n)$ is not periodic because the denominator term is linear in n .

(c) We will show that $x[n]$ is not periodic. Suppose that $x[n]$ is periodic for some period N :

$$e^{j(\frac{\pi}{\sqrt{2}}n)} = e^{j(\frac{\pi}{\sqrt{2}})(n+N)} = e^{j(\frac{\pi}{\sqrt{2}}n+2\pi k)}$$

$$\implies 2\pi k = \frac{\pi}{\sqrt{2}}N, \text{ for integers } k, N$$

$$\implies N = 2\sqrt{2}k, \text{ for some integers } k, N$$

There is no integer k for which N is an integer. Hence $x[n]$ is not periodic.

3. The difference equation:

$$y[n] - \frac{3}{4}y[n-1] + \frac{1}{8}y[n-2] = 2x[n-1]$$

To solve, we take the Fourier transform of both sides.

$$Y(e^{j\omega}) - \frac{3}{4}Y(e^{j\omega})e^{-j\omega} + \frac{1}{8}Y(e^{j\omega})e^{-j2\omega} = 2 \cdot X(e^{j\omega})e^{-j\omega}$$

The system function is given by:

$$\begin{aligned} H(e^{j\omega}) &= \frac{Y(e^{j\omega})}{X(e^{j\omega})} \\ &= \frac{2e^{-j\omega}}{1 - \frac{3}{4}e^{-j\omega} + \frac{1}{8}e^{-j2\omega}} \end{aligned}$$

The impulse response (for $x[n] = \delta[n]$) is the inverse Fourier transform of $H(e^{j\omega})$.

$$H(e^{j\omega}) = \frac{-8}{1 + \frac{1}{4}e^{-j\omega}} + \frac{8}{1 - \frac{1}{2}e^{-j\omega}}$$

Thus,

$$h[n] = -8\left(\frac{1}{4}\right)^n u[n] + 8\left(\frac{1}{2}\right)^n u[n].$$

4. (a)

$$\begin{aligned}
 y[n] &= h[n] * x[n] \\
 &= \sum_{k=-\infty}^{\infty} a^k u[-k-1] u[n-k] \\
 &= \begin{cases} \sum_{k=-\infty}^n a^k, & n \leq -1 \\ \sum_{k=-\infty}^{-1} a^k, & n > -1 \end{cases} \\
 &= \begin{cases} \frac{a^n}{1-1/a}, & n \leq -1 \\ \frac{1/a}{1-1/a}, & n > -1 \end{cases}
 \end{aligned}$$

(b) First, let us define $v[n] = 2^n u[-n-1]$. Then, from part (a), we know that

$$w[n] = u[n] * v[n] = \begin{cases} 2^{n+1}, & n \leq -1 \\ 1, & n > -1 \end{cases}$$

Now,

$$\begin{aligned}
 y[n] &= u[n-4] * v[n] \\
 &= w[n-4] \\
 &= \begin{cases} 2^{n-3}, & n \leq 3 \\ 1, & n > 3 \end{cases}
 \end{aligned}$$

5. Eigenfunctions of LTI systems are of the form α^n , so functions (a) and (b) are eigenfunctions.

Notice that part (c), $\cos(\omega_0 n) = .5(e^{j\omega_0 n} + e^{-j\omega_0 n})$ is a sum of two α^n functions, and is therefore not an eigenfunction itself.

6. (a) No. Consider the following input/outputs:

$$\begin{aligned}
 x_1[n] = \delta[n] &\implies y_1[n] = \left(\frac{1}{4}\right)^n u[n] \\
 x_2[n] = \delta[n-1] &\implies y_2[n] = \left(\frac{1}{4}\right)^{n-1} u[n]
 \end{aligned}$$

Even though $x_2[n] = x_1[n-1]$, $y_2[n] \neq y_1[n-1] = \left(\frac{1}{4}\right)^{n-1} u[n-1]$

(b) No. Consider the input/output pair $x_2[n]$ and $y_2[n]$ above. $x_2[n] = 0$ for $n < 1$, but $y_2[0] \neq 0$.

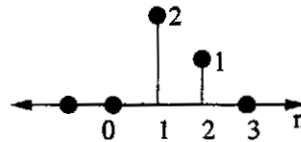
(c) Yes. Since $h[n]$ is stable and multiplication with $u[n]$ will not cause any sequences to become unbounded, the entire system is stable.

7. We use the graphical approach to compute the convolution:

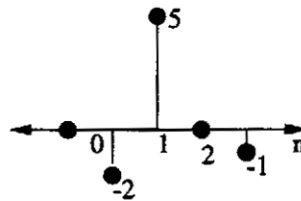
$$\begin{aligned} y[n] &= x[n] * h[n] \\ &= \sum_{k=-\infty}^{\infty} x[k]h[n-k] \end{aligned}$$

(a) $y[n] = x[n] * h[n]$

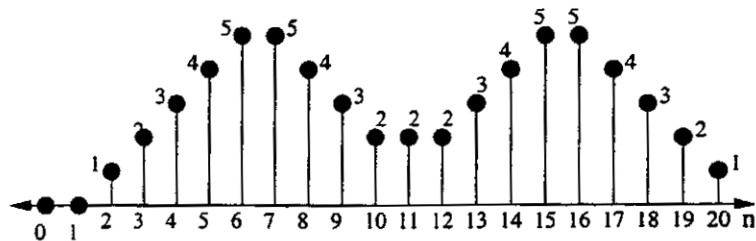
$$y[n] = \delta[n-1] * h[n] = h[n-1]$$



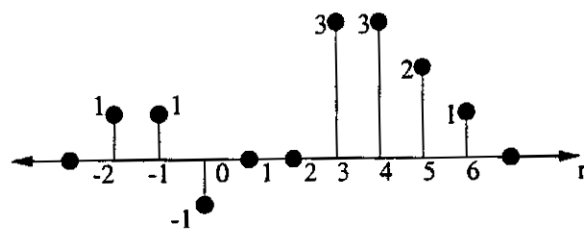
(b) $y[n] = x[n] * h[n]$



(c) $y[n] = x[n] * h[n]$



(d) $y[n] = x[n] * h[n]$



8. The difference equation:

$$y[n] - \frac{1}{2}y[n-1] = x[n] + 2x[n-1] + x[n-2]$$

Taking the Fourier transform of both sides,

$$Y(e^{j\omega})[1 - \frac{1}{2}e^{-j\omega}] = X(e^{j\omega})[1 + 2e^{-j\omega} + e^{-j2\omega}]$$

Hence, the frequency response is

$$\begin{aligned} H(e^{j\omega}) &= \frac{Y(e^{j\omega})}{X(e^{j\omega})} \\ &= \frac{1 + 2e^{-j\omega} + e^{-j2\omega}}{1 - \frac{1}{2}e^{-j\omega}} \end{aligned}$$

9. A system with frequency response:

$$\begin{aligned} H(e^{j\omega}) &= \frac{1 - \frac{1}{2}e^{-j\omega} + e^{-j3\omega}}{1 + \frac{1}{2}e^{-j\omega} + \frac{3}{4}e^{-j2\omega}} \\ &= \frac{Y(e^{j\omega})}{X(e^{j\omega})} \end{aligned}$$

cross multiplying,

$$Y(e^{j\omega})[1 + \frac{1}{2}e^{-j\omega} + \frac{3}{4}e^{-j2\omega}] = X(e^{j\omega})[1 - \frac{1}{2}e^{-j\omega} + e^{-j3\omega}],$$

and the inverse transform gives

$$y[n] + \frac{1}{2}y[n-1] + \frac{3}{4}y[n-2] = x[n] - \frac{1}{2}x[n-1] + x[n-3].$$

EEE401F: Solutions to Problem Set 2

1. Even and odd parts are $x_e[n] = \frac{1}{2}[x(n) + x(-n)]$ and $x_o[n] = \frac{1}{2}[x(n) - x(-n)]$.

a. With $x(n) = u(n)$, we have

$$x_e(n) = \frac{1}{2}[u(n) + u(-n)] = \begin{cases} 1 & n = 0 \\ \frac{1}{2} & n \neq 0 \end{cases}$$

Which may be written as

$$x_e(n) = \frac{1}{2} + \frac{1}{2}\delta(n)$$

Therefore, the even part of the unit step is a sequence that has a constant value of $\frac{1}{2}$ for all n except at $n = 0$, where it has a value of 1. The odd part of the signal $x(n)$ is

$$x_o(n) = \begin{cases} \frac{1}{2} & n > 0 \\ 0 & n = 0 \\ -\frac{1}{2} & n < 0 \end{cases}$$

$$= \frac{1}{2} \text{sgn}(n)$$

where $\text{sgn}(n)$ is the signum function.

b. With $x(n) = \alpha^n u(n)$, the even part is

$$x_e(n) = \frac{1}{2}[\alpha^n u(n) + \alpha^{-n} u(-n)] = \begin{cases} \frac{1}{2}\alpha^n & n > 0 \\ 1 & n = 0 \\ \frac{1}{2}\alpha^{-n} & n < 0 \end{cases}$$

$$= \frac{1}{2}\alpha^{|n|} + \frac{1}{2}\delta(n)$$

The odd part is

$$x_o(n) = \frac{1}{2}[\alpha^n u(n) - \alpha^{-n} u(-n)] = \frac{1}{2}\alpha^{|n|} \text{sgn}(n)$$

2. The conjugate symmetric part of $x(n)$ is

$$x_e(n) = \frac{1}{2}[x(n) + x^*(-n)] = \frac{1}{2}[je^{jn\pi/4} - je^{jn\pi/4}] = 0$$

Thus, this sequence is conjugate antisymmetric.

3. a. The sequence $x(n)$, illustrated in figure 3a, is a linearly decreasing sequence that begins at index $n = 0$ and ends at index $n = 5$. The first sequence that is to be

sketched, $y_1(n) = x(4 - n)$, is found by shifting $x(n)$ by four and time reversing. Observe that at index $n = 4$, $y_1(n)$ is equal to $x(0)$. Therefore, $y_1(n)$ has a value of 6 at $n = 4$ and decreases linearly to the left (decreasing values of n) until $n = -1$, beyond which $y_1(n) = 0$. The sequence $y_1(n)$ is shown in figure 3b.

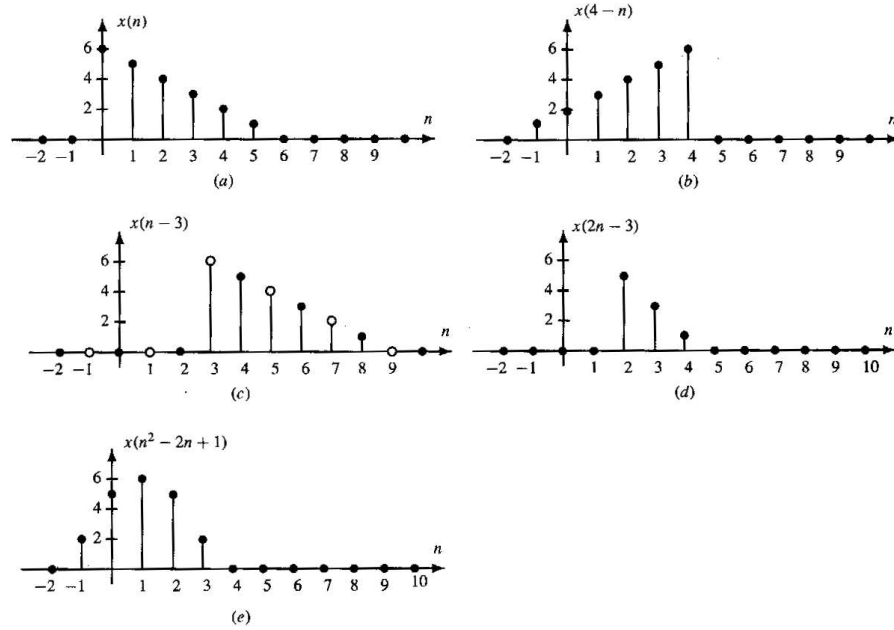


Figure 3.

- b. The second sequence, $y_2(n) = x(2n - 3)$, is formed through the combination of time-shifting and down-sampling. Therefore, $y_2(n)$ may be plotted by first shifting $x(n)$ to the right by three (delays) as shown in figure 3c. The sequence $y_2(n)$ is then formed by down-sampling by a factor of 2 (i.e., keeping only the even index terms as indicated by the solid circles in figure 3c. A sketch of $y_2(n)$ is shown in figure 3d.
- c. Finally, $y_3(n) = x(n^2 - 2n + 1)$ is formed by a nonlinear transformation of the time variable n . This sequence may be easily sketched by listing how the index n is mapped. First, note that if $n \geq 4$ or $n \leq -2$, then $n^2 - 2n + 1 \geq 9$ and, therefore, $y_3(n) = 0$. For $-1 \leq n \leq 3$ we have

$$y_3(-1) = y_3(3) = y_3(4) = 2$$

$$y_3(0) = y_3(2) = x(1) = 5$$

$$y_3(1) = x(0) = 6$$

The sequence $y_3(n)$ is sketched in figure 3g.

4. Note that we may write $x(n)$ as the sum of a left-sided sequence and a right-sided